

ORIGINAL ARTICLE

The United Kingdom Eating Disorders Genetics Initiative

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Abstract

Objective: The United Kingdom Eating Disorders Genetics Initiative (EDGI UK), part of the National Institute for Health and Care Research (NIHR) Mental Health BioResource, aims to deepen our understanding of the environmental and genetic etiology of eating disorders. EDGI UK launched in February 2020 and is partnered with the UK eating disorders charity, Beat. Multiple EDGI branches exist worldwide. This article serves the dual function of providing an in-depth description of our study protocol and of describing our initial sample including demographics, diagnoses, and physical and psychiatric comorbidities.

Method: EDGI UK recruits via media and clinical services. Anyone living in England, at least 16 years old, with a lifetime probable or clinical eating disorder is eligible to sign up online: edgiuk.org. Participants complete online questionnaires, donate a saliva sample for genetic analysis, and consent to medical record linkage and recontact for future studies.

Results: As of September 2022, EDGI UK recruited 7435 survey participants: 98% female, 93.1% white, 97.8% cisgender, 65.9% heterosexual, and 52.1% have a university degree. Over half (57.8%) of these participants have returned their saliva DNA kit. The

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most common diagnoses are anorexia nervosa (48.3%), purging disorder (37.8%), bulimia nervosa (37.5%), binge-eating disorder (15.8%), and atypical anorexia nervosa (7.8%).

Conclusion: EDGI UK is the largest UK eating disorders study and efforts to increase its diversity are underway. It offers a unique opportunity to accelerate eating disorder research. Researchers and participants with lived experience can collaborate on projects with unparalleled sample size.

Public Significance Statement: Eating disorders are debilitating and costly for society but are under-researched due to underfunding. EDGI UK is one of the largest eating disorder studies worldwide with ongoing recruitment. The collected data constitute a resource for secondary analysis. We will combine data from all international EDGI branches and the NIHR BioResource to facilitate research that improves our understanding of eating disorders and their comorbidities.

KEYWORDS

anorexia nervosa, atypical anorexia nervosa, binge-eating disorder, bulimia nervosa, online study, psychiatric genetics, risk factors

1 | INTRODUCTION

1.1 | Eating disorders

Anorexia nervosa, bulimia nervosa, binge-eating disorder, and other specified feeding or eating disorders (OSFED) affect around 8% of the global population, primarily females, with OSFED being the most prevalent (Galmiche et al., 2019). Eating disorders are chronic, with high relapse rates, but only about half of the individuals who develop anorexia or bulimia nervosa reach remission with available treatments (Berkman et al., 2007; Eddy et al., 2017). Stigma, secrecy, and lack of research funding pose barriers to understanding eating disorders fully (Ali et al., 2017; Regan et al., 2017).

Eating disorders have significant costs to healthcare systems and decrease workforce participation, yet many patients' medical needs remain unmet (Ágh et al., 2016; Schaumberg et al., 2017; Streatfeild et al., 2021; Tannous et al., 2021). Moreover, the actual number of affected individuals is underreported as individuals with eating disorder symptoms rarely seek help (Hart et al., 2011). Funding for eating disorder research is disproportionately low taking their prevalence and impact into account. Only ~1% of the UK Government's annual mental health funding was allocated to eating disorder research between 2015 and 2019, despite eating disorders comprising 9% of those with a psychiatric disorder in the United Kingdom (All-Party Parliamentary Group on Eating Disorders, 2021). Eating disorders commonly co-occur with psychiatric and physical illnesses, necessitating increased exploration of shared biological pathways and their impact on therapy outcomes (Eskander et al., 2020; Giel et al., 2022; Gosseume et al., 2019; Momen et al., 2021, 2022; Westmoreland et al., 2016).

1.2 | Genetic factors

Heritability estimates from family and twin studies show a substantial genetic component of anorexia nervosa (58–74%), bulimia nervosa (59–83%), binge-eating disorder (41–57%), and avoidant restrictive food intake disorder (ARFID; 79%) (Dinkler et al., 2022; Thornton et al., 2011). Genome-wide association studies (GWASs) help identify eating disorder-associated common genetic variants that each contribute a small effect to overall risk; however, together they can explain a sizable percentage of trait variance, so-called single-nucleotide polymorphism (SNP)-based heritability. For example, despite the heterogeneity caused by having to include many samples in meta-analyses, attention-deficit/hyperactivity disorder (ADHD) has a SNP-based heritability of 14% (Demontis et al., 2022) and major depressive disorder (MDD) has a SNP-based heritability of 8.9% (Howard et al., 2019; Wray et al., 2018), whilst anorexia nervosa has a SNP-based heritability of 17% (Hübel, Gaspar, Coleman, Finucane, et al., 2019; Hübel, Gaspar, Coleman, Hanscombe, et al., 2019; Watson et al., 2019). Results from GWASs can aid in understanding biological causes and identifying potential drug targets.

Large sample sizes are crucial for GWAS success; international collaboration is necessary to achieve the precision required for identifying the small effects of individual genetic variants. These can be achieved by combining international datasets (Schizophrenia Working Group of the Psychiatric Genomics Consortium, 2014; Sullivan et al., 2018). Several GWASs have successfully identified genomic regions (i.e., loci on chromosomes) associated with psychiatric disorders that commonly co-occur with eating disorders, such as ADHD and MDD (Hübel et al., 2018). For example, 246,363 cases and 561,190 controls were required to identify 101 independent loci

related to depression and MDD (Howard et al., 2019; Wray et al., 2018).

Genetic research in eating disorders lags behind other psychiatric disorders due to funding constraints, resulting in small samples with low statistical power. The Eating Disorders Working Group of the Psychiatric Genomics Consortium (PGC-ED) successfully combined 33 datasets in the largest GWAS of anorexia nervosa (Watson et al., 2019), identifying eight significant loci and genetic correlations with other psychiatric disorders and metabolic traits (Hübel, Gaspar, Coleman, Finucane, et al., 2019; Hübel, Gaspar, Coleman, Hanscombe, et al., 2019). This reconceptualized anorexia nervosa as a metabolic-psychiatric disorder. The Binge Eating Genetics Initiative (BEGIN) study is recruiting individuals to understand the genetics of binge-eating disorder (Bulik et al., 2020). Thus far, no GWASs of clinically diagnosed bulimia nervosa, binge-eating disorder, or ARFID have been conducted, but proxy or machine learned phenotypes have been used (Burstein et al., 2022; Koomar et al., 2021). Well-powered GWASs of all eating disorders in diverse samples are urgently needed to unravel their genetic and biological underpinnings.

1.3 | Environmental and psychological factors

Environmental and psychological factors play a role in the etiology of eating disorders (McClelland et al., 2020; Solmi et al., 2021). Environmental factors such as premature birth (Larsen et al., 2021), older parental age (Larsen et al., 2021), traumatic experiences during early life (Molendijk et al., 2017; Palmisano et al., 2016), and social media use (de Valle et al., 2021), and psychological factors such as low body satisfaction (McLean and Paxton 2019; Stice et al., 2011) have been associated with increased risk. In contrast, positive relationships with parents and family members, including regular family meal times, not commenting on the young person's weight, and body appreciation can be protective against eating disorder development (Langdon-Daly & Serpell, 2017). Behaviors such as dietary restriction, self-induced vomiting, or laxative intake often precede binge-eating disorder and bulimia nervosa (Allen et al., 2013). As longitudinal studies are rare, it is difficult to distinguish prodromal symptoms from risk factors. Understanding how these factors interact with genetic vulnerability is still not fully understood.

1.4 | Aims of EDGI UK

The United Kingdom Eating Disorders Genetics Initiative (EDGI UK; edgiuk.org; Figure 1a) aims to enhance our understanding of eating disorders by collecting genetic and phenotypic data from at least 10,000 people and linking it with their medical records. Furthermore, EDGI UK is a volunteer resource as part of the National Institute for Health and Care Research (NIHR) BioResource—a UK-based database of >250,000 volunteers—enabling secondary analysis. Internationally, EDGI UK is part of a research collaboration, EDGI World (Bulik et al., 2021), with teams in the United States, Australia, New Zealand, Sweden, Puerto Rico, and Mexico. EDGI World aims to collect 100,000 genetic samples of each eating disorder. As of 2023, more countries

(e.g., Italy) are opening EDGI branches. EDGI UK represents a novel and vital resource for research in eating disorders and associated comorbidities. This article serves the dual function of providing an in-depth description of our study protocol and of describing our initial sample including demographics, diagnoses, and physical and psychiatric comorbidities. We focus on the most frequently-reported diagnoses in our sample: anorexia nervosa, atypical anorexia nervosa, bulimia nervosa, binge-eating disorder, and purging disorder.

2 | METHODS

2.1 | Pre-launch

2.1.1 | Collaboration with EDGI World and Beat

EDGI World is the successor to the Anorexia Nervosa Genetics Initiative (ANGI; Figure 1b). The UK branch of EDGI pioneered a detailed

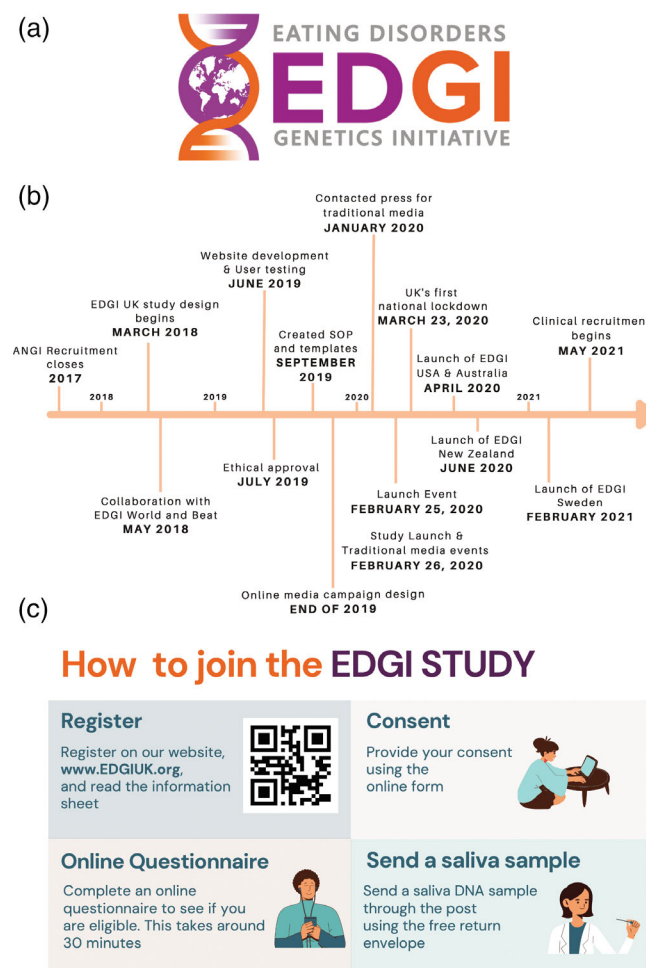


FIGURE 1 Panel a: The United Kingdom Eating Disorders Genetics Initiative (EDGI UK) logo. Panel b: Timeline of EDGI UK. The timeline begins at the end of the Anorexia Nervosa Genetics Initiative (ANGI) in 2017 and ends with the writing of this manuscript in September 2022. Panel c: Sign-up process of EDGI UK. The four-step sign-up process for participants of EDGI UK highlights registration, consent, completing the online questionnaire, and the return of their saliva sample.

phenotyping approach, which was then adopted by other branches worldwide and tailored to their specific needs. EDGI UK formed a partnership with Beat, a UK eating disorder charity (beateatingdisorders.org.uk). Beat's involvement as part of the UK's All-Party Parliamentary Group for Eating Disorders, consisting of government members and experts working on improving policies, was invaluable. Beat plays a vital role in providing helplines, raising awareness, disseminating information, and promoting research projects like EDGI UK. The collaboration includes participation in Beat's annual conference during Eating Disorders Awareness Week and featuring EDGI UK on their social media, website, and mailing lists. Beat's ambassadors, who have personal experience with eating disorders or care for someone with an eating disorder, contributed to promotional materials and offered insights as part of EDGI UK's Steering Committee.

2.1.2 | Website development

EDGI UK's participation process was conducted through edgiuk.org. The website was based on the Genetic Links to Anxiety and Depression (GLAD) Study (Davies et al., 2019; gladstudy.org.uk), and the color scheme highlights the partnership with Beat, who provided input on the website design. For a £10 Amazon voucher, Beat ambassadors and volunteers tested participant-facing materials, website design, and functionality. Their feedback enhanced the user experience.

2.1.3 | Standard operating procedures and templates

Standard operating procedures were established for participant contact, database management, saliva kit posting, and record-keeping. Templates for email, social media, and voicemail responses were created to ensure consistent and timely communication with participants, providing information on the study, website usage, saliva sample procedures, and an overview of genetics.

2.1.4 | Endorsement through organizations and individuals

Prior to the launch, EDGI UK sought endorsements from organizations, charities, politicians, and mental health activists to advertise on their public platforms. An information package was sent, outlining recruitment support strategies such as spokesperson roles and social media quotes. Mental health organizations like F.E.A.S.T, Body Dysmorphic Disorder Foundation, MQ Mental Health, The Motherhood Group, and Mental Health Foundation endorsed EDGI UK. Influencers, athletes, and mental health activists provided further support and endorsements on the one-year anniversary of EDGI UK. Notable activists include Hope Virgo and Dave Chawner, who serve as spokespeople for the initiative.

2.2 | Launch event

We organized an EDGI UK launch event, which included members of the eating disorder community, National Health Service (NHS) staff, university representatives, the NIHR BioResource, and Beat. The event featured presentations by Beat, the NIHR BioResource, EDGI UK's chief investigators, and individuals with lived experience, emphasizing the significance of our research. The event concluded with a panel discussion and a question-and-answer session.

2.3 | Traditional media campaign

To secure traditional media coverage, Bright Star Digital, the NIHR BioResource, and the King's College London Communications and Engagement teams reached out to newspapers and radio stations. Sky News conducted an interview with Professor Gerome Breen and Hope Virgo. Professor Janet Treasure and Andrew Radford, Beat's Chief Executive, appeared on BBC Radio 4 Woman's Hour. Pre-recorded content by Professor Gerome Breen was played on 10 regional radio stations (recordings are only available for purchase due to copyright restrictions). Tom Quinn, the Director of External Affairs for Beat, spoke to ITV News, the UK's largest commercial news organization. Online articles were published by ITV News, Pharma Times, the NIHR BioResource, and Beat (Supplementary Material 1). Media coverage fell short of expectations due to the overwhelming focus on COVID-19 reporting, as EDGI UK launched just three weeks before the UK Government declared the first national COVID-19 lockdown in March 2020.

2.4 | Social media campaign

EDGI UK utilized various online advertising strategies, including paid social media ads developed and distributed by Born Social (bornsocial.com) on Instagram® and Facebook®. These ads targeted young people of all genders and emphasized the inclusive nature of EDGI UK, welcoming participation regardless of eating disorder type, gender, or clinical diagnosis. The first ad encompassed all eating disorders, while the second specifically focused on the underdiagnosed and overlooked binge-eating disorder (Kornstein et al., 2016). To align with Beat's media values, animated body-image neutral figurines or diverse body types were featured. Initially, trigger warnings were included in the ad captions but later removed due to insufficient evidence of their efficacy (Sanson et al., 2019) and concerns about perpetuating stigma (edgiuk.org/trigger-warning-statement). Instagram users were given the option to opt-out of seeing our paid ads on their Instagram® feeds.

In addition to the agency-created content, we created a six-week social media content plan for Instagram®, Facebook®, and Twitter®. Posts included quotes from EDGI UK spokespeople, infographics on EDGI UK and eating disorders. They also carried Beat's, EDGI UK's, and the NIHR BioResource's logos to showcase our collaborations. We pinned an in-house produced "Call to

Action” video to the top of our Twitter® page: lived experience volunteers spoke about their eating disorders and our chief investigators highlighted the urgent research need and explained the participation process.

2.5 | Clinical recruitment

NHS England Trusts are hospital and community services that provide patient care in circumscribed geographical regions. By April 2023, EDGI UK recruited 248 participants through 18 of England's NHS trusts via specialist First Episode Rapid Early Intervention for Eating Disorders (FREED) services (Richards et al., 2022). Clinical recruitment was delayed until early 2021 due to COVID-19 restrictions and availability of medical staff. Each trust displays posters, distributes leaflets, sends texts or letters to patients with eating disorders, or approaches patients in-clinic to assist with registration and completing questionnaires.

2.6 | Eligibility criteria

To participate in EDGI UK, individuals must be 16 years or older, reside in England, and have a lifetime experience of an eating disorder according to the Diagnostic and Statistical Manual of Mental Health, Fifth Edition (DSM-5; American Psychiatric Association, 2013) or the International Classification of Diseases, Eleventh Revision World Health Organisation (ICD-11; WHO, 2019). Clinical diagnosis or treatment is not necessary, as eligibility is determined through the sign-up questionnaire. Current funding and ethical approval limitations restrict EDGI UK to England, although individuals living abroad who have an English home address (e.g., university students) are eligible. Ineligible individuals with self-reported lifetime anxiety and/or depressive disorders are invited to the GLAD Study (gladstudy.org.uk). While approximately half of EDGI UK survey participants have submitted saliva samples for genotyping, this article includes all participants who completed the sign-up questionnaire, regardless of sample return (i.e., EDGI UK survey participants).

2.7 | Sign-up process

Interested individuals can watch an animated video introducing EDGI UK and the NIHR BioResource prior to enrolment. To sign up on edgiuk.org, participants create an account, provide their name, contact information, sex, gender, and date of birth. Subsequently, they must review participant information and provide electronic consent. Mandatory consent for EDGI UK includes joining the NIHR BioResource, allowing long-term de-personalized data storage, donating a biological sample, being contacted for future research projects based on genetic/phenotypic data, receiving additional questionnaires, and permitting linkage of medical records. After consenting, participants complete the EDGI UK sign-up questionnaire, which determines eligibility. Eligible participants receive a saliva kit by mail for further analysis (Figure 1c).

2.8 | Measures

The online assessment consists of two components: one sign-up questionnaire and multiple optional questionnaires. First, the sign-up questionnaire assesses basic demographics, psychiatric and physical comorbidity, and family history via the NIHR BioResource Health and Lifestyle questionnaire. The ED100K questionnaire assesses lifetime eating disorder symptoms, their severity, and duration (Thornton et al., 2018). The ED100K is augmented by questions covering atypical anorexia nervosa, the validated 7-Item Binge Eating Disorder Screener (BEDS-7; Herman et al., 2016), the Night Eating Diagnostic Questionnaire (NEDQ; Gluck et al., 2001), the Muscle Dysmorphic Disorder Inventory (MDDI; Hildebrandt et al., 2004), and the Nine-Item Avoidant/Restrictive Food Intake Disorder Screen (NIAS; Zickgraf & Ellis, 2018) (Supplementary Material 2). Participants can skip the Childhood Trauma Screener (CTS-5; Glaesmer et al., 2013) and the Post-Traumatic Stress Disorder Checklist (PCL-6; Lang et al., 2012), regarding adverse life experiences including domestic violence and catastrophic trauma (Davis et al., 2020). The sign-up questionnaire takes ~30–45 min depending on routing. Second, participants can access 32 optional questionnaires on symptomatology, comorbidity, treatment, and social impairment in six groups: eating behaviors and physical activity, thoughts and behaviors, lifestyle, quality of life, personality, and treatment (for all measures, see Supplementary Material 3). Our assessment battery mirrors large-scale data collections: the UK Biobank (Davis et al., 2020), two Swedish eating disorder quality registers (Stepwise and Riksät; Birgegård et al., 2010), and EDGI World.

2.9 | Biological samples, genotyping, genetic analysis

We send eligible participants a package including a GFX Isohelix Saliva Collector with instructions (Supplementary Material 4) for self-sampling at home. Participants can return their samples free of charge by post. To

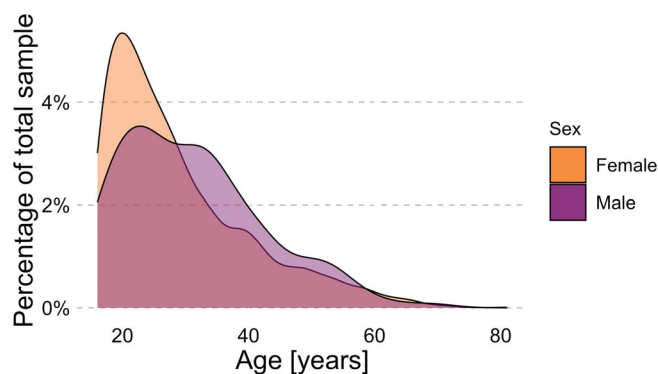


FIGURE 2 Age distribution of sample. Density plot illustrating the age distribution of the Eating Disorders Genetics Initiative United Kingdom (EDGI UK) survey participants ($n = 7435$) stratified by sex assigned at birth (female $n = 7286$; male $n = 149$), as of September 2022.

TABLE 1 Characteristics of the Eating Disorders Genetics Initiative United Kingdom (EDGI UK) survey participants.

	Total (n = 7435)	Anorexia nervosa (n = 3590)	Atypical anorexia nervosa (n = 582)	Binge-eating disorder (n = 1176)	Bulimia nervosa (n = 2787)	Purging disorder (n = 2813)
Female	7286 (98.0%)	3525 (98.2%)	563 (96.7%)	1151 (97.9%)	2731 (98.0%)	2774 (98.6%)
Age						
Mean (SD)	29 (11)	27 (10)	28 (12)	35 (12)	29 (11)	27 (10)
Median (IQR)	26 (20, 35)	23 (19, 30)	24 (20, 33)	34 (26, 42)	27 (21, 35)	24 (20, 31)
Range	16–81	16–80	16–76	16–76	16–72	16–80
Gender						
Man	177 (2.4%)	76 (2.1%)	19 (3.3%)	32 (2.7%)	70 (2.5%)	53 (1.9%)
Non-binary	205 (2.8%)	95 (2.7%)	26 (4.5%)	42 (3.6%)	85 (3.0%)	90 (3.2%)
Prefer to self-define	39 (0.5%)	17 (0.5%)	<5 (<0.9%)	10 (0.9%)	15 (0.5%)	18 (0.6%)
Woman	6975 (94.0%)	3386 (94.5%)	528 (90.9%)	1085 (92.3%)	2606 (93.5%)	2638 (93.9%)
Transgender	167 (2.2%)	80 (2.2%)	22 (3.8%)	30 (2.6%)	71 (2.5%)	72 (2.6%)
Sexual orientation						
Asexual	125 (1.7%)	84 (2.3%)	9 (1.5%)	37 (3.1%)	39 (1.4%)	46 (1.6%)
Bisexual	1623 (21.8%)	753 (21.0%)	136 (23.4%)	264 (22.4%)	694 (24.9%)	687 (24.4%)
Heterosexual	4898 (65.9%)	2386 (66.5%)	361 (62.0%)	731 (62.2%)	1751 (62.8%)	1755 (62.4%)
Homosexual	362 (4.9%)	171 (4.8%)	31 (5.3%)	58 (4.9%)	144 (5.2%)	174 (6.2%)
Other	295 (4.0%)	124 (3.5%)	35 (6.0%)	70 (6.0%)	128 (4.6%)	119 (4.2%)
Ethnic origin						
Arab	14 (0.2%)	9 (0.3%)	<5 (<0.9%)	<5 (<0.4%)	9 (0.3%)	7 (0.2%)
Asian or Asian British	116 (1.6%)	62 (1.7%)	5 (0.9%)	14 (1.2%)	46 (1.7%)	51 (1.8%)
Black or Black British	27 (0.4%)	10 (0.3%)	<5 (<0.9%)	8 (0.7%)	11 (0.4%)	13 (0.5%)
Mixed	261 (3.5%)	133 (3.7%)	23 (4.0%)	35 (3.0%)	113 (4.1%)	126 (4.5%)
Other	82 (1.1%)	42 (1.2%)	6 (1.0%)	18 (1.5%)	41 (1.5%)	35 (1.2%)
White	6922 (93.1%)	3327 (92.7%)	544 (93.5%)	1097 (93.3%)	2561 (91.9%)	2577 (91.6%)
Highest education level						
None listed	117 (1.6%)	64 (1.8%)	14 (2.4%)	11 (0.9%)	35 (1.3%)	45 (1.6%)
CSEs or equivalent	47 (0.6%)	28 (0.8%)	5 (0.9%)	6 (0.5%)	22 (0.8%)	26 (0.9%)
O-levels/GCSEs or equivalent	885 (12.0%)	493 (13.8%)	76 (13.2%)	103 (8.8%)	283 (10.2%)	365 (13.0%)
NVQ/HND/HNC or equivalent	264 (3.6%)	79 (2.2%)	22 (3.8%)	55 (4.7%)	100 (3.6%)	91 (3.3%)
AS-levels/A-levels or equivalent	2209 (30.0%)	1191 (33.3%)	181 (31.4%)	269 (23.1%)	794 (28.7%)	899 (32.1%)
University degree or equivalent	3840 (52.1%)	1713 (47.9%)	278 (48.2%)	720 (61.9%)	1534 (55.4%)	1368 (48.9%)
Current employment status						
Doing unpaid or voluntary work	87 (1.2%)	48 (1.3%)	9 (1.5%)	10 (0.9%)	22 (0.8%)	27 (1.0%)
Full-time or part-time student	1998 (26.9%)	1206 (33.6%)	156 (26.8%)	160 (13.7%)	676 (24.3%)	816 (29.0%)
In paid employment or self-employed	3934 (53.0%)	1599 (44.6%)	278 (47.8%)	739 (63.1%)	1570 (56.3%)	1415 (50.3%)
Looking after home and/or family	128 (1.7%)	41 (1.1%)	6 (1.0%)	46 (3.9%)	42 (1.5%)	45 (1.6%)
Other	196 (2.6%)	108 (3.0%)	12 (2.1%)	27 (2.3%)	80 (2.9%)	84 (3.0%)
Retired	85 (1.1%)	24 (0.7%)	12 (2.1%)	20 (1.7%)	24 (0.9%)	12 (0.4%)

TABLE 1 (Continued)

	Total (n = 7435)	Anorexia nervosa (n = 3590)	Atypical anorexia nervosa (n = 582)	Binge-eating disorder (n = 1176)	Bulimia nervosa (n = 2787)	Purging disorder (n = 2813)
Unable to work because of sickness or disability	611 (8.2%)	355 (9.9%)	75 (12.9%)	131 (11.2%)	226 (8.1%)	271 (9.6%)
Unemployed	368 (5.0%)	196 (5.5%)	32 (5.5%)	37 (3.2%)	143 (5.1%)	141 (5.0%)
<i>Current marital/relationship status</i>						
Divorced	183 (2.5%)	51 (1.4%)	16 (2.7%)	56 (4.8%)	59 (2.1%)	49 (1.7%)
Married/civil partnership	1155 (15.5%)	371 (10.3%)	89 (15.3%)	299 (25.4%)	447 (16.0%)	362 (12.9%)
Other	29 (0.4%)	17 (0.5%)	<5 (<0.9%)	5 (0.4%)	11 (0.4%)	15 (0.5%)
Relationship (living together)	1380 (18.6%)	599 (16.7%)	109 (18.7%)	249 (21.2%)	581 (20.8%)	516 (18.3%)
Relationship (not living together)	1380 (18.6%)	682 (19.0%)	107 (18.4%)	156 (13.3%)	536 (19.2%)	591 (21.0%)
Separated	91 (1.2%)	33 (0.9%)	8 (1.4%)	33 (2.8%)	31 (1.1%)	28 (1.0%)
Single	3192 (42.9%)	1822 (50.8%)	250 (43.0%)	374 (31.8%)	1117 (40.1%)	1242 (44.2%)
Widowed	19 (0.3%)	8 (0.2%)	0 (0.0%)	<5 (<0.4%)	5 (0.2%)	5 (0.2%)
<i>Smoking: pack years</i>						
Mean (SD)	4.9 (7.7)	3.8 (6.4)	4.1 (6.2)	7.8 (10.9)	4.6 (6.8)	3.9 (6.3)
Median (IQR)	2.0 (0.5, 6.0)	1.3 (0.4, 4.4)	2.0 (0.4, 4.5)	3.8 (1.0, 10.0)	2.0 (0.5, 5.4)	1.5 (0.4, 4.5)
Range	0.0–90.0	0.0–60.0	0.0–47.5	0.0–90.0	0.0–55.0	0.0–55.0
Received treatment	5062 (68.7%)	3012 (85.3%)	398 (68.4%)	514 (43.9%)	1659 (59.9%)	2040 (72.9%)

Note: The total sample ($n = 7435$) and each subsample defined by diagnosis: anorexia nervosa (AN; $n = 3590$), atypical anorexia nervosa (AAN; $n = 582$), binge-eating disorder (BED; $n = 1176$), bulimia nervosa (BN; $n = 2787$), and purging disorder (PUR; $n = 2813$), September 2022. Diagnoses are self-reported via the mental health diagnosis (MHD) questionnaire and/or derived via The Diagnostic and Statistical Manual of Mental Disorders, Fifth Edition (DSM-5) diagnostic algorithms based on responses to the ED100K. Percentages are based on complete data, that is, those who skipped the question or responded with “Prefer not to answer” or “Do not know” are not included in the calculation. Participants could be included in more than one eating disorder group.

Abbreviations: A level, advanced level; AS level, advanced subsidiary level; CSE, Certificate of Secondary Education; GCSE, General Certificate of Secondary Education; HNC, Higher National Certificates; HND, Higher National Diplomas; IQR, interquartile range; n , number; NVQ, National Vocational Qualification; SD, standard deviation.

ensure return of the kit, participants receive up to six emails and two text message reminders, along with tips for providing a saliva sample. Participants are responsible for posting their samples to the National Biosample Centre in Milton Keynes. The DNA is extracted following the manufacturer's instructions and stored via the NIHR BioResource at the National Biosample Centre in Milton Keynes. Genotyping will be conducted on the UK Biobank Axiom microarray with subsequent quality control. Through the international PGC-ED collaboration, we will perform GWAS meta-analyses combining EDGI UK genotypes with data from the UK Biobank, EDGI World, and the Lundbeck Foundation Initiative for Integrative Psychiatric Research, Denmark.

2.10 | Participant recontact and access to EDGI UK

Researchers can either submit proposals for participant recontact or access de-identified data for their own research projects. During the

COVID-19 pandemic, participants were invited to the COVID-19 Psychiatry and Neurological Genetics (COPING) study, which began at the first UK lockdown to monitor pandemic effects on mental health in which >900 EDGI UK survey participants took part (Davies et al., 2022; Young et al., 2021). As of 2022, EDGI UK is part of the UK Longitudinal Linkage Collaboration (UK LLC), a national collaborative effort for harmonizing health data for longitudinal research.

2.11 | Statistical analysis

Here, we included participants with age, sex, and reported lifetime experience of an eating disorder ($n = 7435$). For continuous variables, we calculated means, standard deviations, medians, and interquartile ranges and show minima and maxima. For categorical variables, we present counts and percentages of the total sample unless otherwise stated. We derived diagnoses applying DSM-5-based algorithms to ED100K responses or participants self-reported a diagnosis via the

Mental Health Diagnosis (MHD) questionnaire (Davis et al., 2020; Thornton et al., 2018). In our analyses, some eating disorders are algorithm-derived only (i.e., anorexia nervosa binge-eating/purging and anorexia nervosa restricting) whilst others are self-report only (i.e., OSFED, pica, rumination disorder, and ARFID). Medical record linkage between EDGI UK and records from services such as primary and secondary care, the mental health service use dataset, and hospital episode statistics is underway and will enable the verification of questionnaire based diagnoses. Psychiatric comorbidities are self-reported via the MHD questionnaire and physical comorbidities are self-reported via the NIHR BioResource's Health and Lifestyle questionnaire. Code for all analyses including algorithms can be found here: https://github.com/tnggroup/EDGI_protocol.

3 | RESULTS

3.1 | Sample descriptors

Recruitment of EDGI UK is ongoing and results are from participants recruited before September 16, 2022. In the first week of recruitment, 2236 participants signed up to EDGI UK and 780 of these (35%) completed the sign-up questionnaire. Within the first month, 4389 signed up and 1537 (36%) completed the sign-up questionnaire. As of September 16, 2022, 15,339 people registered, 10,319 (67%) consented to participate, 7435 (48.5%) completed the sign-up questionnaire (i.e., EDGI UK survey participants) and were included in our report, and 4294 (28%) returned their saliva kits for genotyping. We recruited 5758 participants (~77% of eligible individuals) via social media who heard about the study on Facebook, Twitter®, or Instagram. By September 2022, @edgi_uk had over 4000 Instagram followers. Most @edgi_uk followers are women (96%), aged between 25 and 34 years, and based in the United Kingdom, predominantly London. On Twitter, @edgi_uk has 1014 followers, and on Facebook, @edgi.uk has 814. On the latter platforms, no detailed information is available.

EDGI UK survey participants ($n = 7435$) are on average 29 years old (Figure 2), mostly female (98%), cisgender (97.8%), white (93.1%), heterosexual (65.9%), have a university degree or equivalent (52.1%), and are in paid employment (53%; Table 1). The five most common lifetime eating disorders reported are anorexia nervosa (48.4% of females, 43.6% of males), atypical anorexia nervosa (7.7% of females, 12.8% of males), bulimia nervosa (37.5% of females, 37.6% of males), binge-eating disorder (15.8% of females, 16.8% of males), and purging disorder (38.1% of females, 26.2% of males; Table 2). Eating disorder diagnoses in EDGI UK show a large degree of overlap (Figure 3). Other lifetime eating disorders present in EDGI UK include atypical binge-eating disorder (0.6% of sample), atypical bulimia nervosa (1.7%), night-eating syndrome (1.4%), pica (0.2%), ARFID (1.6%), rumination disorder (0.2%), and OSFED/EDNOS (3.1%; Table 2). Demographic characteristics such as age, sexual orientation, and ethnicity are largely consistent across the whole sample and each of the five most common eating disorder groups (Table 1). Over half of participants report that they have received treatment at some point in their life (68.7%; Table 1). Treatment receipt was

TABLE 2 Lifetime self-reported and/or Diagnostic and Statistical Manual of Mental Disorders, Fifth edition (DSM-5) algorithm-derived eating disorder diagnoses in the Eating Disorders Genetics Initiative United Kingdom (EDGI UK) survey participants split by biological sex as of September 2022.

	Female ($n = 7286$)	Male ($n = 149$)
<i>Anorexia nervosa</i>		
Binge-eating/purging ^a	2071 (28.4%)	38 (25.5%)
Restricting ^a	1149 (15.8%)	20 (13.4%)
Unknown subtype	305 (4.2%)	7 (4.7%)
<i>Bulimia nervosa</i>	2731 (37.5%)	56 (37.6%)
Binge-eating disorder	1151 (15.8%)	25 (16.8%)
Purging disorder	2774 (38.1%)	39 (26.2%)
Atypical anorexia nervosa	563 (7.7%)	19 (12.8%)
Atypical binge-eating disorder	42 (0.6%)	<5 (<4.0%)
Atypical bulimia nervosa	121 (1.7%)	<5 (<4.0%)
Night-eating syndrome	101 (1.4%)	<5 (<4.0%)
Pica ^b	12 (0.2%)	0 (0.0%)
Avoidant restrictive food intake disorder ^b	117 (1.6%)	<5 (<4.0%)
Rumination disorder ^b	11 (0.2%)	0 (0.0%)
OSFED/EDNOS ^b	213 (3.1%)	<5 (<4.0%)

Note: Diagnoses are self-reported via the Mental Health Diagnosis (MHD) questionnaire and/or derived by applying Diagnostic and Statistical Manual of Mental Disorders, Fifth Edition (DSM-5) diagnostic algorithms to responses to the ED100K. Sex: assigned sex at birth; n : number; OSFED/EDNOS: other specified feeding and eating disorders/eating disorder not otherwise specified.

^aDiagnoses are algorithm-derived only.

^bDiagnoses are self-report only. Participants could be assigned to more than one eating disorder group.

most often reported by those with lifetime anorexia nervosa (85.3%), and least often by those with binge-eating disorder (43.9%).

3.2 | Psychiatric comorbidities

Over half of EDGI UK survey participants report a lifetime diagnosis of depression and anxiety, with the highest proportion in those with binge-eating disorder (67%; Table 3). Other highly endorsed psychiatric comorbidities include obsessive compulsive disorder/s (OCD; 24.2–33.4%), posttraumatic stress disorder (15–20.8%), and personality disorder (13–25%). More participants with binge-eating disorder endorsed ADHD (11.2%), bipolar disorder (8.5%), and a personality disorder (18.1%), than people with other eating disorders such as anorexia nervosa, with 5.2%, 4.5%, and 13.0%, respectively.

3.3 | Physical comorbidities

Approximately half of EDGI UK survey participants report an allergy (43.6–55.4%), and ~25% report a lung and/or breathing illness (21–

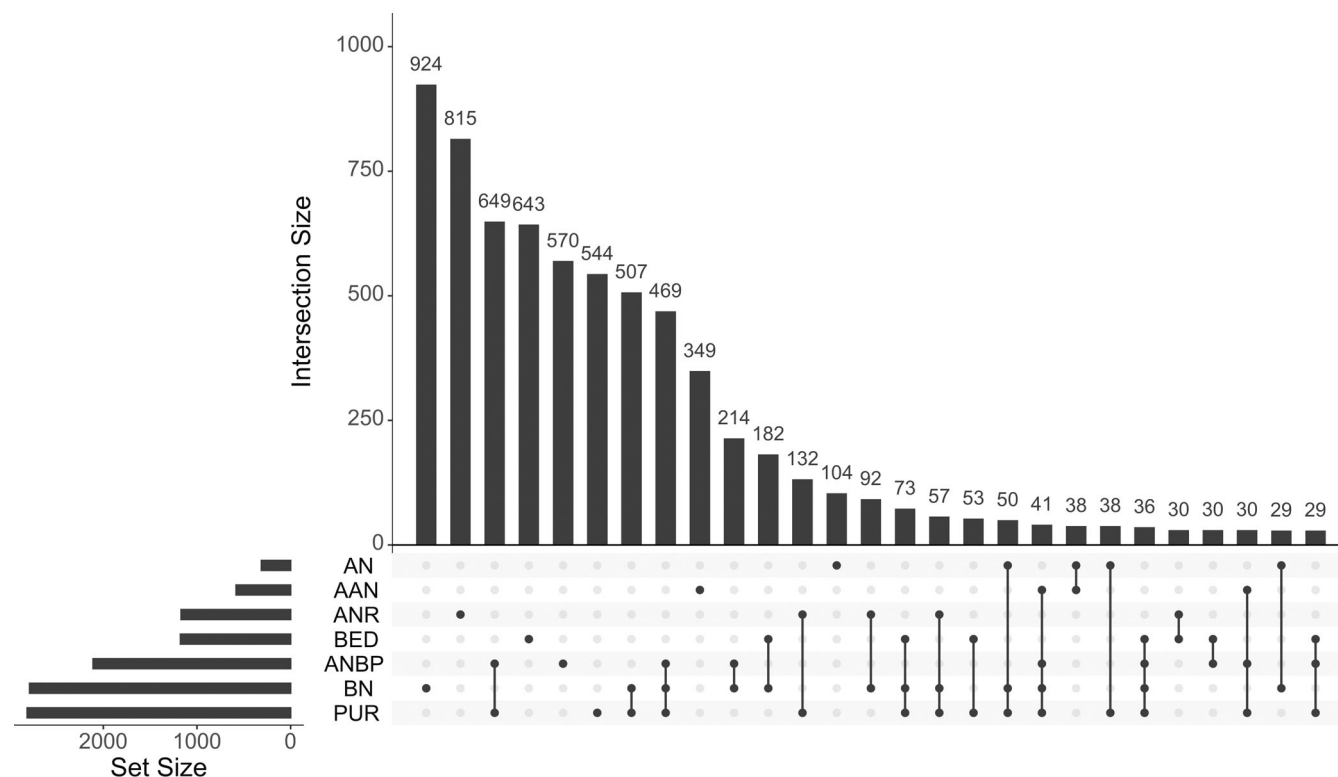


FIGURE 3 Overlap of lifetime eating disorder diagnoses in Eating Disorders Genetics Initiative United Kingdom (EDGI UK) survey participants. Diagnoses are algorithm-derived based on responses to the ED100K and/or self-report in the mental health diagnosis (MHD) questionnaire. Some eating disorders are algorithm-derived only (anorexia nervosa binge-eating/purging and anorexia nervosa restricting). Set size represents the overall size of each eating disorder group, in which people with multiple eating disorder diagnoses will be counted more than once. Intersection size indicates size of overlapping groups: the filled-in circles beneath each intersection-size column indicate which eating disorder/s are part of that intersection. Where multiple eating disorders are part of an intersection, a line is drawn between their corresponding filled-in circles. Groups smaller than 20 are not presented. See Supplementary Figure 1 for more information on other groups. AAN; atypical anorexia nervosa; AN, anorexia nervosa (unknown subtype); ANBP, anorexia nervosa binge-eating/purging; ANR, anorexia nervosa restricting; BED, binge-eating disorder; BN, bulimia nervosa; PUR, purging disorder.

31%; Table 4). Diabetes mellitus (6.5%), polycystic ovary syndrome (15.3%), and thyroid disease (9.9%) are most highly endorsed in those with binge-eating disorder.

3.4 | Completed research and policy work

Based on data from EDGI UK, we have finalized several reports. In collaboration with Beat, we published a report titled “When do eating disorders start? An investigation into two large UK samples,” which revealed that low weight, purging, and binge eating start as frequently over the age of 18 as under the age of 18 (Davies et al., 2021). Professor Gerome Breen and Dave Chawner discussed these findings on Sky News, emphasizing the need for increased investment in adult NHS eating disorder services (Parliamentary and Health Service Ombudsman, 2017). From early 2021, Professor Gerome Breen and other experts have repeatedly met with the UK Government to advocate for changes in eating disorder research and treatment policies. Initial findings from the COPING study, which included a subset of EDGI UK survey participants, indicated that individuals with psychiatric disorders and higher levels of pandemic-related concerns were more likely to experience new-onset binge eating

or low weight during the pandemic (Davies et al., 2022). Other findings suggested that being female, younger, or unemployed/student was associated with worsening depression, anxiety, or post-traumatic stress disorder (PTSD) symptoms during the pandemic (Young et al., 2021). A network analysis of symptoms reported by EDGI UK participants found more consistent differences in the presence and structure of mania symptoms when comparing those with and without binge eating than those seen between people with different binge-type eating disorders (Davies et al., 2023).

4 | DISCUSSION

Despite the impact of COVID-19, EDGI UK stayed on track with its recruitment targets, with 7435 participants completing the sign-up questionnaire and 4294 returning their saliva kits by September 2022. Common lifetime diagnoses included anorexia nervosa, atypical anorexia nervosa, binge-eating disorder, bulimia nervosa, and purging disorder. It is common for people to experience more than one eating disorder in their lifetime, which our data demonstrate. In genetic studies, researchers can study “broad” phenotypes in which more lenient

TABLE 3 Psychiatric comorbidities in the Eating Disorders Genetics Initiative United Kingdom (EDGI UK) survey participants.

	Anorexia nervosa (n = 3590)	Atypical anorexia nervosa (n = 582)	Binge-eating disorder (n = 1176)	Bulimia nervosa (n = 2787)	Purging disorder (n = 2813)
<i>Depression and anxiety</i>					
No depressive or anxiety disorder	634 (18.9%)	101 (18.4%)	125 (10.9%)	454 (17.2%)	395 (14.9%)
Depressive and anxiety disorder	1833 (54.7%)	323 (58.9%)	766 (67.0%)	1521 (57.7%)	1617 (61.1%)
Only depressive disorder	521 (15.5%)	80 (14.6%)	185 (16.2%)	432 (16.4%)	403 (15.2%)
Only anxiety disorder	363 (10.8%)	44 (8.0%)	68 (5.9%)	228 (8.7%)	232 (8.8%)
PTSD	539 (15.0%)	107 (18.4%)	245 (20.8%)	438 (15.7%)	508 (18.1%)
Obsessive compulsive disorder/s	1110 (33.1%)	167 (30.4%)	277 (24.2%)	741 (28.1%)	885 (33.4%)
Personality disorder	466 (13.0%)	145 (25.0%)	212 (18.1%)	416 (15.0%)	433 (15.4%)
<i>Bipolar disorder and psychotic disorder</i>					
No psychotic or bipolar disorder	3084 (93.0%)	501 (92.4%)	1007 (89.0%)	2392 (91.9%)	2401 (91.3%)
Psychotic and bipolar disorder	40 (1.2%)	12 (2.2%)	19 (1.7%)	28 (1.1%)	38 (1.4%)
Only psychotic disorder	84 (2.5%)	13 (2.4%)	28 (2.5%)	60 (2.3%)	73 (2.8%)
Only bipolar disorder	109 (3.3%)	16 (3.0%)	77 (6.8%)	123 (4.7%)	117 (4.5%)
ASD	245 (6.9%)	75 (13.0%)	70 (6.0%)	140 (5.1%)	167 (6.0%)
ADHD	185 (5.2%)	50 (8.6%)	132 (11.2%)	175 (6.3%)	155 (5.5%)

Note: Each subsample is defined by diagnosis: anorexia nervosa (n = 3590), atypical anorexia nervosa (n = 582), binge-eating disorder (n = 1176), bulimia nervosa (n = 2787), and purging disorder (n = 2813), as of September 2022. Psychiatric comorbidities are self-reported via the mental health diagnosis (MHD) questionnaire. Percentages are based on complete data, that is, those who skipped the question or responded with “Prefer not to answer” or “Do not know” are not included in the calculation. Participants could be assigned to more than one eating disorder group. “Personality disorder” includes: Paranoid, schizoid, schizotypal, antisocial, borderline/emotionally unstable, histrionic, narcissistic, avoidant/anxious, dependent, obsessive-compulsive, and/or mixed personality disorders. “Psychotic disorders” include schizophrenia and any other type of psychosis or psychotic illness. Depressive disorders include major depression, antenatal/postnatal depression, premenstrual depression. Anxiety disorders include generalized anxiety, social anxiety, specific phobia, agoraphobia, panic disorder, and panic attacks.

Abbreviations: ADHD, attention deficit hyperactivity disorder; ASD, autism spectrum disorder; PTSD, post-traumatic stress disorder.

inclusion criteria are applied, for instance, including people with multiple eating disorder diagnoses in a study of a single eating disorder, or studying a single symptom such as binge eating. Contrastingly, researchers can also apply stricter criteria and study “narrow” phenotypes, for example, including only those with a single diagnosis of the eating disorder under study. Comorbid psychiatric and physical illnesses, such as depression, anxiety disorders, OCD, and PTSD, were prevalent. Allergies, dermatological, and pulmonological diseases were the most reported physical comorbidities. Social media played a significant role in recruitment, and the online nature of the study allowed participants to conveniently participate from home, especially during the pandemic. EDGI UK is well-positioned to create a substantial resource for future collaboration in eating disorders research.

The assessment battery in EDGI UK draws from the Swedish eating disorder quality registers Stepwise and RIKSÄT (Birgegård et al., 2010; Groß et al., 2013), in which symptoms are evaluated through clinical interviews and jointly registered by the patient and medical professional online. These registers also track long-term clinical outcomes included in genetic studies (Johansson et al., 2022). Similarly, EDGI UK assesses and tracks clinical outcomes. Unlike EDGI UK, Stepwise focuses on individuals for whom an intention to treat has been established, limiting the sample to clinical cases. However, the percentages of registered individuals with anorexia (27%–40%) and bulimia nervosa (9%–29%) in Stepwise align with those in EDGI UK (Birgegård et al., 2010).

The distribution of eating disorders in EDGI UK differs from the general population (Micali et al., 2017). Epidemiological studies suggest that binge-eating disorder is more prevalent than anorexia nervosa (Kessler et al., 2013). The GLAD Study (Davies et al., 2019), our sister project, recruited more people with binge-eating disorder, likely due to their experience of comorbid depression. Participants with comorbid depression and binge-eating disorder may be more aware of their depression, motivating them to join the GLAD Study instead of EDGI UK. Stigma, shame (Ali et al., 2017; Hepworth & Paxton, 2007), and the lack of recognition of binge-eating disorder as a psychiatric disorder amongst the general public (Reas, 2017) may also deter individuals from joining EDGI UK. The launch of the GLAD Study prior to EDGI UK may have contributed to this recruitment pattern. However, both EDGI UK and the GLAD Study are part of the NIHR BioResource and share phenotype assessment hence allowing for combined analysis.

4.1 | Limitations

EDGI UK is a selected sample that comprises far more females than the general English population (98% vs. 51%) (Population and Household Estimates, England and Wales: Census 2021, 2021). This likely reflects recruitment biases, as it is suggested that ~25% of

TABLE 4 Physical comorbidities (disorders, diseases, or symptoms) in the Eating Disorders Genetics Initiative United Kingdom (EDGI UK) survey participants.

	Anorexia nervosa (n = 3590)	Atypical anorexia nervosa (n = 582)	Binge-eating disorder (n = 1176)	Bulimia nervosa (n = 2787)	Purging disorder (n = 2813)
Cancer	13 (0.4%)	<5 (<0.9%)	11 (0.9%)	14 (0.5%)	9 (0.3%)
Neurological	764 (21.3%)	152 (26.1%)	381 (32.4%)	711 (25.5%)	710 (25.2%)
Allergies	1566 (43.6%)	286 (49.1%)	651 (55.4%)	1310 (47.0%)	1345 (47.8%)
Musculoskeletal	840 (23.4%)	120 (20.6%)	273 (23.2%)	490 (17.6%)	580 (20.6%)
Lung and breathing illnesses	753 (21.0%)	163 (28.0%)	364 (31.0%)	700 (25.1%)	734 (26.1%)
Cardiovascular	228 (6.4%)	54 (9.3%)	248 (21.1%)	258 (9.3%)	223 (7.9%)
Gastrointestinal	101 (2.8%)	14 (2.4%)	35 (3.0%)	61 (2.2%)	76 (2.7%)
Diabetes mellitus	39 (1.1%)	10 (1.7%)	77 (6.5%)	59 (2.1%)	45 (1.6%)
Polycystic ovary syndrome	220 (6.1%)	43 (7.4%)	180 (15.3%)	263 (9.4%)	203 (7.2%)
Dermatological	1128 (31.4%)	188 (32.3%)	434 (36.9%)	918 (32.9%)	929 (33.0%)
Thyroid disease	195 (5.4%)	33 (5.7%)	116 (9.9%)	174 (6.2%)	157 (5.6%)

Note: Each subsample is defined by diagnosis: anorexia nervosa (n = 3590), atypical anorexia nervosa (n = 582), binge-eating disorder (n = 1176), bulimia nervosa (n = 2787), and purging disorder (n = 2813), as of September 2022. Physical comorbidities are self-reported via the National Institute for Health and Care Research (NIHR) BioResource's Health and Lifestyle questionnaire. Percentages are based on complete data, that is, those who skipped the question or responded with "Prefer not to answer" or "Do not know" are not included in the calculation. Participants could be assigned to more than one eating disorder group. "Cancer" includes any lung, breast, stomach, colon, uterus and/or prostate cancer; "Neurological" includes epilepsy or convulsions, migraines, multiple sclerosis, Parkinson's disease, brain tumor, and/or severe memory loss; "Allergies" includes hay fever, drug allergy, food allergy, and/or other allergy; "Musculoskeletal" includes ankylosing spondylitis, osteoporosis, osteoarthritis, hyper-mobility, rheumatoid arthritis, and/or other arthritis; "Lung and breathing illnesses" includes asthma and/or emphysema or chronic bronchitis; "Cardiovascular" includes heart attack or angina, high blood cholesterol, high blood pressure, atrial fibrillation, stroke, and/or paroxysmal tachycardia syndrome (PoTS); "Gastrointestinal" includes Crohn's disease, ulcerative colitis, and/or coeliac disease; "Diabetes mellitus" includes diabetes type 1 (early and late onset), diabetes type 2 (late onset), and/or pain due to diabetes (diabetic neuropathy); "Dermatological" includes psoriasis, eczema, vitiligo, and/or post herpetic neuralgia. See Supplementary Table 1 for comorbidity frequencies of each individual physical illness.

individuals with an eating disorder are male (Sweeting et al., 2015). One factor complicating recruitment is that males often do not recognize that they have an eating disorder and hence are undiagnosed (Strother et al., 2012). Moreover, EDGI UK is less diverse than the English and Welsh general population (93% vs. 84% white) (Population Estimates by Ethnic Group and Religion, England and Wales 2109, 2019), limiting our investigations in minoritized ethnic and racial groups.

Internet access is necessary to take part, which may pose a barrier for individuals who lack the financial means or computer literacy. Participants without internet access are advised to visit their local library, ask a trusted family member or a friend, or visit a participating NHS Trust. The reliance on social media for advertising may result in older individuals having less exposure to study promotions.

Thus far, the ED100K, which assesses eligibility, has only been validated for anorexia nervosa in US-American and Australian samples (Thornton et al., 2018). We have successfully obtained ethical approval at King's College London to further validate the ED100K questionnaire in inpatients and outpatients. Moreover, the NIAS, used to assess ARFID symptoms, does not include items measuring malnutrition, general practitioner (GP) visits, or social impairment, complicating the assignment of an algorithm-derived ARFID diagnosis. To overcome this limitation, we intend to include the Pica, ARFID, and Rumination Disorder Interview Questionnaire (PARDI-AR-Q; Bryant-Waugh et al., 2019) for participants who meet the cut-off of 24 points on the NIAS.

4.2 | Future directions

EDGI UK has a clear roadmap for improvement. To begin with, we will enhance racial and ethnic diversity in our community by organizing public and patient involvement events, engaging in outreach, and employing tailored advertising. We will broaden the demographics of our participants by including individuals under the age of 16 and extending our recruitment efforts to cover Northern Ireland, Scotland, and Wales, which is targetted in 2023/2024. We will employ targeted advertising and outreach strategies to increase male participation. We plan to collaborate with additional NHS Trusts to enhance our clinical recruitment. Ultimately, EDGI UK is an exceptional resource that substantially contributes to international genetics research in eating disorders. With our goal of recruiting 10,000 participants, we are at the forefront of understanding the genetic and environmental influences on eating disorders.

AUTHOR CONTRIBUTIONS

Dina Monsсен: Conceptualization; data curation; investigation; methodology; project administration; resources; visualization; writing – original draft; writing – review and editing. **Helena L. Davies:** Conceptualization; data curation; formal analysis; investigation; methodology; project administration; software; validation; visualization; writing – original draft; writing – review and editing. **Saakshi Kakar:** Data curation; formal analysis; resources; software; validation; writing – review and editing. **Shannon Bristow:** Data curation; formal analysis; funding acquisition;

project administration; resources; software; validation; visualization; writing – review and editing. **Steven Bright:** Formal analysis; writing – review and editing. **Suzannah C. B. Curzons:** Writing – review and editing. **Molly R. Davies:** Data curation; project administration; resources; writing – review and editing. **Emily J. Kelly:** Data curation; resources; writing – review and editing. **Jessica Mundy:** Data curation; software; validation; writing – review and editing. **Alicia J. Peel:** Data curation; software; validation; writing – review and editing. **Gursharan Kalsi:** Data curation; funding acquisition; project administration; resources; writing – review and editing. **Chelsea Mika Malouf:** Data curation; resources; writing – review and editing. **Alish B. Palmos:** Data curation; software; validation; writing – review and editing. **Henry C. Rogers:** Data curation; resources; writing – review and editing. **Abigail R. Ter Kuile:** Data curation; software; validation; writing – review and editing. **Sang Hyuck Lee:** Writing – review and editing. **Johan Zvrskovec:** Data curation; software; validation; writing – review and editing. **Zain Ahmad:** Data curation; software; validation; writing – review and editing. **Jonathan R. I. Coleman:** Data curation; software; validation; writing – review and editing. **Matthew Hotopf:** Funding acquisition; writing – review and editing. **Megan Skelton:** Writing – review and editing. **Brett N. Adey:** Writing – review and editing. **Hope Virgo:** Conceptualization; writing – review and editing. **Kiran Glen:** Data curation; resources; writing – review and editing. **Tom Price:** Writing – review and editing. **John R. Bradley:** Writing – review and editing. **Tom Quinn:** Conceptualization; writing – review and editing. **Nathalie Kingston:** Writing – review and editing. **Monika McAtarsney-Kovacs:** Writing – review and editing. **Thalia C. Eley:** Funding acquisition; writing – review and editing. **Janet Treasure:** Conceptualization; funding acquisition; project administration; supervision; writing – review and editing. **Christopher Hübel:** Conceptualization; data curation; formal analysis; investigation; methodology; project administration; resources; software; supervision; validation; visualization; writing – original draft; writing – review and editing. **Gerome Breen:** Conceptualization; funding acquisition; investigation; methodology; project administration; resources; supervision; writing – review and editing.

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CONFLICT OF INTEREST STATEMENT

Prof. Gerome Breen has received honoraria, research or conference grants and consulting fees from Illumina, Otsuka, and COMPASS Pathfinder Ltd. Prof. Matthew Hotopf is principal investigator of the RADAR-CNS consortium, an IMI public private partnership, and as such receives research funding from Janssen, UCB, Biogen, Lundbeck and MSD.

DATA AVAILABILITY STATEMENT

EDGI UK data are not publicly available however are available via a data request application to the NIHR BioResource (<https://bioresource.nihr.ac.uk/using-our-bioresource/academic-and-clinical-researchers/apply-for-bioresource-data/>). All code for this study is publicly available: https://github.com/tnggroup/EDGI_protocol.

ETHICS STATEMENT

EDGI UK was approved by the London – Fulham Research Ethics Committee on July 29, 2019 (REC reference: 19/LO/1254) following a full review. The NIHR BioResource has been approved as a Research Tissue Bank by the East of England – Cambridge Central Committee (REC reference: 17/EE/0025).

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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